1.4 Modern and smart materials, composite materials and technical textiles

1.4.1 Modern and smart materials

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Smart materials change or react to a change in their environment, such as temperature, light, pressure or electrical input. Reactions include a change in colour, shape or resistance.

Shape-memory alloys

- Shape-memory alloys (SMAs) appear to have a memory that allows them to return to an original shape if heated.
- Nitinol, an alloy of nickel and titanium, is a commonly used SMA. Possible uses include medical applications such as medical fastenings used in bone fractures.

Nanomaterials

- Nanomaterials are defined as having particles on nanoscale dimensions, approximately equivalent to one billionth of a metre.
- In textiles, nanoparticles can alter the properties of a fabric, for example adding a stain-resistant finish.
- New materials have been developed as a result of miniaturisation of component parts, which has led to the development of **conductive fibres**, commonly referred to as **e-textiles**.
- Conductive fibres and threads developed from carbon, steel and silver can be woven into textile fabrics and made into clothing, or the conductive threads can be sewn into a product to connect a circuit. This allows interaction between the fabric and the user.

Reactive glass

- Reactive glass reacts almost immediately to bright sunlight. It is controlled by a **light-dependent resistor (LDR)**.
- Smart glass reacts in a similar way but has to be activated by a switch. It is opaque when switched off. It is used in homes to give privacy instead of using curtains or blinds.

Piezoelectric materials

- Piezoelectric materials create an electrical charge when exposed to compression or pressure. Possible uses include sound or sonar detection and lighters for grills and barbeques.

Temperature-responsive polymers

- Photochromic pigments or dyes change colour in response to changes in light. For example, sunglasses can change colour in response to UV radiation.
- Thermochromic pigments or dyes change colour in response to a change in heat and can be engineered to specific heat ranges.

Conductive fibres: fibres that conduct electricity.

E-textiles: textile fabrics that have integrated and interactive devices built into the fabric that react with the user.

Light-dependent resistor (LDR): a sensor input that can detect changes in light levels.



Figure 1.6 Flexible solar panels can now be integrated into textiles fabrics



Figure 1.7 Thermochromic mugs change colour when boiling water is poured in

Conductive inks

Conductive inks contain conductive particles that allow designers to hand-draw or digitally print circuits that are fully functional.

1.4.2 Composites

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Composites are created when two or more materials are joined together to create a new, enhanced material. One material is known as the **matrix** while the other is the **reinforcement**.

Composite material: a new material created by combining two or more different types of material.

Compressive strength: the ability to withstand a pushing (squashing) force.

Concrete

- Concrete is made up of various aggregates (gravel, rock or sand), cement and water. The cement binds the materials together when water is added. Concrete has a high **compressive strength** but is much weaker under tension.
- Uses include the construction of buildings and roads, as well as smaller projects such as kerbs and drainage.

Plywood

- Plywood is made up of several layers of wood glued on top of each other. Each layer is laid at a 90-degree angle to the previous layer, giving plywood a consistent strength. The outside layers might be a more expensive wood such as birch.
- Uses include roofing, flooring and furniture.



Figure 1.8 Plywood

Fibre/carbon/glass and reinforced polymers

- Carbon fibre reinforced polymer (CFRP) is a composite consisting of woven carbon fibre strands encased in a polymer resin.
- Carbon fibre strands have a high tensile strength and the polymer resin is lightweight and rigid; when combined, this creates a high-performance engineered material. Possible uses include high-performance sports equipment.
- Glass reinforced plastic (GRP) is a composite of glass fibres and a polyester resin. Glass fibres create rigidity and the resin makes GRP tough and lightweight.
- GRP is difficult to recycle because the process of combining the glass fibres and resin cannot be reversed.

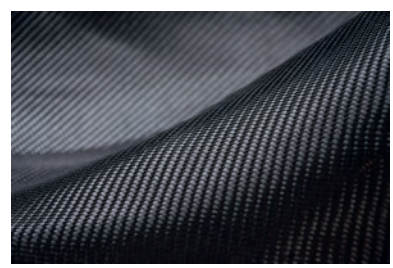


Figure 1.9 CFRP matting

Robotic materials

- Scientists are developing materials that will appear to 'think' for themselves. It is anticipated that these materials will sense something, process it, then react to that sensation, although this is some way off from being a reality.
- Possible uses could include camouflage or load balancing.

1.4.3 Technical textiles

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- Technical textiles are engineered with specific performance characteristics that suit a particular purpose or function.
- Technical textiles can be woven or bonded, synthetic or natural.

Table 1.8 Technical textiles and their uses

Textile	Where used	Uses
Agro-textiles	Agricultural industry	Mainly used in crop protection to suppress weeds or for moisture control
Construction textiles	Construction industry	Netting used on scaffolding to protect passers-by from falling debris; linings for swimming pools and ponds
Geotextiles	Civil engineering, road construction and building	Permeable fabrics used with soil that have the ability to filter, separate, protect and drain; control of embankments on the sides of roads and protection to prevent coastal erosion

Domestic textiles

- Domestic textiles have various applications in the home, from padding inside soft furnishing to cleaning cloths.
- **Microfibres** are engineered fibres that are about 100 times smaller than a human hair. When used in cleaning clothes they are highly effective in attracting dust.

Microfibres: tiny fibres about 100 times thinner than a human hair.

Environmentally friendly textiles

- Environmentally friendly textiles come from sustainable sources such as plants and animals.
- However, to be considered truly eco-friendly they have to be grown organically, without the use of pesticides and fertilisers to boost crops.

Protective textiles

- Gore-Tex is the most well-known breathable fabric and consists of three or more fabrics laminated together with a breathable **hydrophilic membrane** in the middle:
  - Warm air and tiny droplets of moisture from perspiration can permeate out through the breathable membrane but moisture from larger rain droplets and wind cannot enter.
  - It is used in high performance clothing and footwear, and it helps to regulate body temperature.
- **Aramid fibres** are engineered fibres with very high tensile strength and resistance to heat.

Hydrophilic membrane: a solid structure that stops water passing through but, at the same time, can absorb and diffuse fine water vapour molecules.

Aramid fibres: engineered synthetic polymers that are very strong and resistant to heat.

Exam tip

Be able to fully explain how smart, composite and technical textiles are used in the design and manufacture of products. Know their specific properties and what makes each suitable for a particular purpose in a product.

Typical mistake

Don't confuse technical textiles such as Gore-Tex with fabrics such as cotton that absorb moisture making them appear breathable.

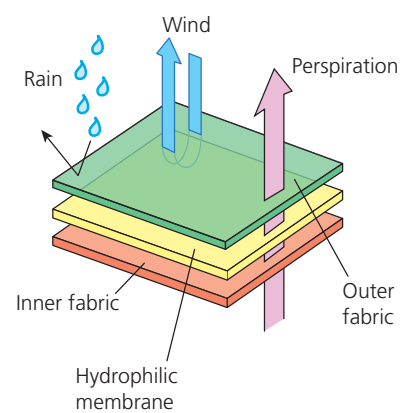


Figure 1.10 Gore-Tex is an example of a laminated fabric

Table 1.9 Aramid fibres

Fibre	Properties	Uses
Kevlar®	Lightweight, flexible and extremely durable aramid; excellent resistance to heat, corrosion and damage from chemicals and high tensile strength-to-weight ratio	Used in protective clothing such as police body armour, where the fibre is woven in a lattice that provides protection against knife attack
Nomex®	Extremely strong synthetic fabric that can withstand exposure to the most extreme conditions	Primarily used where resistance to heat and flames is essential, for example firefighters' uniforms

Sports textiles

- **Micro-encapsulation** is a process of applying microscopic capsules to fibres and fabrics.
- The capsules can contain vitamins, therapeutic oils, moisturisers, antiseptics and antibacterial chemicals that are released through friction.
- Rhovyl® is a non-flammable, synthetic fibre that is crease resistant, has good thermal and acoustic properties, is antibacterial and comfortable to wear.
- The construction of the fibre gives fabrics the ability to wick away moisture, such as perspiration, through the fabric. It also dries quickly, meaning it does not retain odours, making it ideal for socks and sportswear.

Micro-encapsulation: tiny microscopic droplets containing various substances applied to fibres, yarns and materials including paper and card.

Now test yourself

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1 Describe what gives plywood its strength. (2)
2 Explain why carbon fibre reinforced polymer (CFRP) is a suitable material for a racing bike. (3)
3 Explain how micro-encapsulation is beneficial to a patient when used in medical dressings. (2)
4 State what is meant by 'smart glass'. (1)
5 Describe **two** reasons why the aramid fibre Nomex® is used in protective clothing. (2)

1.5 Mechanical devices

1.5.1 Types of movement

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The four main types of movement are:

- **linear**: movement in a straight line
- **reciprocation**: movement that goes back and forth in a straight line
- **rotary**: movement that turns in a circle
- **oscillation**: a swinging movement from side to side.

Durability

- **Durability** is the wood's ability to resist decay from exposure to the elements and attack from insects.
- Hardwoods such as teak are the most durable woods and can last for many years when exposed to the elements.
- Some types of bamboo have a durability of only a few months.

Durability: resistance to wear, long lasting.

Now test yourself

TESTED

1 Why is plywood a much stronger manufactured board than MDF? (2)
2 Name **one** advantage and **one** disadvantage of knots in pine. (4)
3 Explain **one** major difference between hardwood and softwood trees. (2)

1.13 Contexts which inform design outcomes

1.13.1 Discriminating between materials, components and manufacturing processes

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Properties may change depending on things such as temperature or the shape and direction of the material. Other properties will remain the same regardless of the environment or conditions.

Mechanical properties include:

- compressive strength – resistance to a pushing force
- tensile strength – resistance to a pulling force
- durability – resistance to wear, long lasting
- hardness – resistance to abrasion
- stiffness – resistance to bending and flexing
- elasticity – ability to stretch and return to shape
- impact resistance – resistance to knocks and impacts
- brittleness – hard but easily broken or shattered
- water resistance – resistance to water or moisture
- absorbency – ability to soak up moisture.

Manufacturing properties include:

- machinability – ability to be easily milled and drilled
- workability – ability to be placed and compacted
- ductility – ability to stretch, not break under stress or impact
- malleability – ability to be hammered, pressed or rolled without breaking.

Other properties include:

- acoustic – acoustic absorption
- chemical – corrosion resistance
- electrical – conductivity, resistance and so on
- thermal – conductivity, flash point and so on
- magnetic – attractive to magnets.

Typical mistake

Malleability and ductility are similar properties and often confused. Ductility relates to the ability of a material to stretch without damage. Malleability relates to the ability of a material to deform under compression.

Exam tip

By learning the properties of materials, you can understand why they have been used to make certain products. This will help you to answer questions in the exam about individual products and their fitness for purpose.

Advantages and disadvantages

- The choice of materials impacts the success of the final product.
- The advantages of a particular material may make it an obvious choice for a product, but the designer must also consider disadvantages, for example a thermosetting polymer may have the best properties for a product but it is not recyclable. Therefore, the designer may choose a thermoforming polymer with properties as close to the thermosetting one as possible.

Justification of the choice

- The designer must make informed choices based on these characteristics and other factors, such as:
  - cost – is there another material available that may cost less?
  - availability – is the material in good supply?
  - sustainability – is the material sustainable?
  - environmental impact – can the material be recycled?
  - social, cultural and moral considerations – will the use of this material have a negative impact?
- When designing an electronic system, the designer must consider the range of components available and decide on the most suitable type for the application.

Now test yourself

TESTED

1 State **four** important mechanical properties of materials. (4)
2 State **two** factors that can influence and affect the properties of materials. (2)

1.14 Environmental, social and economic challenges

1.14.1 Respect for different social, ethnic and economic groups

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Positive and negative impacts of products

- Products can have a negative impact on society that the designer may not have foreseen.
- For example, leaflets and flyers can be used to communicate useful information to people about services in the area, however this can lead to unwanted junk mail being delivered, urging people to buy things they don't want.
- While it is not always easy to predict the effects of a product, designers must try to consider the possible negative implications and decide whether the product is worthwhile.